Hardy Wang

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TECHNICAL SKILLS

Board Design (Altium, LTSpice): schematic capture and layout of a steering wheel board (**Rev 3.0**), an RF sensor node, and a **4-layer radar board**
Embedded Firmware (C/C++, Python, STM32, ESP32): wrote the firmware for an FM band sweep over **I2C and SPI**, STM32 board with **CAN, UART, and I2C**
Bring-up & Rework (THT/SMD Soldering): hand assembled and reworked boards, a custom PCB that **worked on first bring-up** with cleaner readings than the breadboard prototype
Bench Testing (Oscilloscope, Multimeter, Function Generator, VNA): measured output ripple, **VNA S-parameter measurements**, traced an **ADC fault** to an ADC and WiFi SoC conflict
Power & Signal Integrity: replaced an end of life buck converter and bench validated it, opto-isolation, return paths, ground pour cutouts

EDUCATION

University of British Columbia **Vancouver, BC**
Bachelor of Applied Science in Electrical Engineering *Expected graduation: Dec. 2028*
Available for internships: May to August 2027

EXPERIENCE

Hardware Engineering Intern **May 2026 – Present**
e-Radio Inc. *Toronto, ON*

- Took a two-layer RF sensor board from **schematic through layout, fab, and bring-up**, and wrote the ESP32 firmware that runs on it, including the **I2C and SPI** drivers.
- Tuned the receiver's per-channel settle time from measured data and added a warm-up dwell to remove a **first-sweep reading error**, so every sweep is usable.
- Built the data log to **survive power loss** and sped up uploads with batching so the backlog **no longer grows**.
- Traced an **ADC fault** to an ADC and WiFi SoC resource conflict, fixed it in firmware, and **avoided a board respin**. Used **VNA S-parameter measurements** to develop the RF front end.

Power and Signals Designer **Sept 2025 – Present**
UBC Solar Car *Vancouver, BC*

- One of **~8 engineers** on the Power and Signals team. Designed the steering wheel board (**7 driver inputs** over CAN, UART, and I2C) from schematic to a race-ready unit, syncing weekly with mechanical and software subteams on connectors, mounting, and firmware interfaces.
- Selected components against the team budget, tracked issues and action items on my board and teammates' boards, and hand assembled and bench validated prototype and final boards.

Project Technician Intern **May 2026 – Present**
Aecon Group Inc. *Toronto, ON*

- Reviewed electrical Bills of Materials against engineering design standards, **catching component mismatches** before they reached procurement.

PROJECTS

Steering Wheel PCB | Altium Designer, STM32F103 **Dec 2025 – Apr 2026**

- High-transient signals near the MCU risked corrupting driver input logic (horn, push-to-talk, regenerative braking). I split the problem into noise sources, return paths, and sensitive nets, then **isolated them with opto-isolators, dedicated return paths, and ground pour cutouts**, and verified clean operation on the bench.
- Routed **CAN, UART, and I2C** off the STM32F103RCT6 for **7 driver inputs**, and migrated programming and debug from ST-Link to **JTAG**, verifying pin routing on the bench.

Embedded RF Sensor Node | Altium Designer, ESP32, Si4732, C++ **May 2026 – Aug 2026**

- Sole hardware designer of an FM band monitor (ESP32, Si4732 receiver, DS3231 RTC, microSD, NTC thermistor) built as **20 units** through JLCPCB. Strapped the receiver's bus-mode pins with **10k pull-up and pull-down resistors** so it latches I2C at reset, and placed **0 ohm jumpers** so the ESP32 could drive the reference clock, which **recovered a failed crystal without a respin**.
- Firmware: a **watchdog**, a boot-loop counter that drops the unit into **safe mode**, an **interrupt-driven debounced button**, thermistor readings with hysteresis and **out-of-range fault rejection**, a **CRC-16 on every SD record**, **I2C bus recovery** at boot, and a serial **self-test with a PASS/FAIL board verdict**.

Automotive Radar Perception Node | Altium Designer, STM32H7 **Sept 2026 – Present**

- Designing a **4-layer radar board** around a BGT24MTR11 and STM32H743VIT6 ($150 CAD budget). Chose the tuning and monitor filters (**16 Hz and 1.6 kHz**) and **ferrite filtered VDDA/VREF+** to keep switching noise out of the ADC reference, and defined the IF chain to be sized from chirp time and ADC range.